Univerza v Mariboru
Ekonomsko-poslovna fakulteta
Razlagova 14
2000 Maribor, Slovenija

Guidelines on the Use of Generative Artificial Intelligence In Pedagogical and Research Activities at University of Maribor Faculty of Economics and Business (UM FEB)

1. Introduction

The use of artificial intelligence (AI), particularly generative tools such as ChatGPT, Copilot, and others, introduces numerous new opportunities in educational and research environments. In the field of economics and business, these tools provide support for pedagogical activities, professional work, and research. However, it is essential that their use aligns with the principles of responsibility, ethics, and academic integrity.

This document employs the following terms:

- **Foundations:** Fundamental points or premises from which decisions and actions derive.
- **Principles:** General guidelines or basic rules that direct actions and decisions.
- **Guidelines:** Specific recommendations or instructions detailing how certain activities should be carried out.

2. Foundations for the Use of Generative AI at UM FEB

The following foundational points apply to the use of generative AI in pedagogical and research processes at UM FEB:

- Higher education teachers and students must be adequately informed about the capabilities and limitations of these tools.
- AI should act as a supportive tool, not a replacement for human judgment and critical thinking.
- Understanding ethical aspects such as content generation impartiality, data security, and privacy protection is crucial.
- Irresponsible use of AI can lead to misinformation or plagiarism, making adherence to academic standards imperative.

3. Use of Generative AI in Pedagogical Processes

The principles guiding the integration of AI in pedagogical processes are based on practices from leading educational institutions, such as Harvard University (Harvard University, 2023). Depending on the course's nature and instructor expectations, UM EPF may follow one of three fundamental principles:

i) Prohibition of AI Use: Instructors may require that all student-submitted work be original and free of generative AI tools. In such cases, all use of AI, including during preparatory stages, is explicitly prohibited.

Guidelines for providing instructions to students:

"We expect all work submitted for this course, 'COURSE NAME,' to be original and authored by the student. For group assignments, we expect all members who contributed to the task to be named. The use of tools such as ChatGPT, Copilot, or any other generative AI tools is explicitly prohibited during all stages of work, including preparatory stages. Note that principles regarding AI use may vary between courses, and students are responsible for familiarizing themselves with AI use policies for each specific course."

ii) Encouraging AI Use: Instructors may encourage using generative AI tools, such as ChatGPT, as research tools. However, students must clearly acknowledge and cite their use of such tools in their work.

Guidelines for providing instructions to students:

"This course, 'COURSE NAME,' encourages students to explore generative AI tools, such as ChatGPT, for assignments and evaluations. Any use of these tools must be fully disclosed and cited appropriately. Students are responsible for verifying the accuracy and validity of all results generated by AI; the final responsibility for submitted work lies with the student. Note that principles regarding AI use may vary between courses, and students are responsible for familiarizing themselves with AI use policies for each specific course."

iii) Shared Principles: Instructors may permit the use of generative AI tools for certain tasks while explicitly prohibiting them for others. This approach offers flexibility based on the task's nature and course objectives.

Guidelines for providing instructions to students:

"Certain assignments for this course, 'COURSE NAME,' may allow or even encourage the use of generative AI tools, such as ChatGPT, Copilot, etc. Generally, the use of such tools is not permitted unless explicitly stated otherwise. Any use of generative AI tools must be appropriately disclosed and cited. Students are responsible for ensuring the validity and appropriateness of all results generated by AI; the final responsibility for submitted work lies with the student. Violations of these principles will be treated as breaches of academic integrity. Note that principles regarding AI use may vary between courses, and students are responsible for familiarizing themselves with AI use policies for each specific course."

4. Use of Generative AI for Student Work Evaluation

Principle: Responsible use of generative AI for evaluating student work must adhere to principles of privacy protection, intellectual property respect, and transparency.

When considering the appropriateness of using AI for providing feedback on student work, each higher education instructor must carefully evaluate whether it is suitable to input student work into AI systems, such as ChatGPT, Copilot, or similar tools, for checking originality or other aspects. This decision must be made cautiously, taking into account the potential impacts on privacy and intellectual property.

Guideline: Confidential or sensitive data must not be entered into generative AI systems, as these systems do not ensure privacy or confidentiality.

Teachers must obtain explicit, documented consent from students before submitting original student work to AI tools for feedback or assessment.

- **Guideline:** Teachers must secure explicit, documented consent before entering student work into AI tools.

Students should also be informed about the risks associated with submitting their work to AI systems, including the possibility of the provider accessing, storing, or using their data.

- **Guideline:** The use of AI for originality checks or assessment must always be transparent, and students should be informed of associated risks.

5. Use of Generative AI in Research

Research processes involving generative AI require additional attention to ensure research quality, ethics, and transparency.

Key principles for research (European Commission, 2024):

- **Reliability:** Researchers must ensure the quality of their research by using verified sources and carefully reviewing AI-generated results.
- **Fairness:** Any use of AI in the research process must be explicitly disclosed and cited.
- **Respect:** AI use must respect participant privacy, intellectual property rights, and applicable laws.
- **Responsibility:** Researchers are accountable for results supported by AI tools, monitoring potential biases and errors from AI use.

Guidelines for responsible AI use in research (European Commission, 2024):

- Researchers must use AI tools transparently and in accordance with applicable laws, including national, European, and international legislation.
- AI systems should not be used in situations that could involve the unauthorized disclosure of confidential information related to human research, unpublished studies, potential intellectual property claims, or proprietary and classified research.
- Researchers bear full responsibility for the integrity of content created with or supported by AI tools. AI systems cannot be considered authors or co-authors. Authorship implies action and responsibility, which remain within the domain of human researchers.

Recommendations for Researchers (Resnik and Hosseini, 2024; European Commission, 2024):

- Researchers should disclose, describe, and explain their use of AI tools in research, including any limitations, in language that is accessible to non-experts. References to the tool should include its name, version, date of use, and additional details about how it was employed and how it influenced the research process. Researchers may include relevant prompts used in the system and the resulting outputs, in line with principles of open science.
- Particular attention must be given to privacy issues and intellectual property rights when handling data generated by AI tools.
- Researchers must continually educate themselves on AI usage and the ethical issues that arise from its application.
- Researchers should identify, describe, mitigate, and monitor biases and random errors associated with AI.
- Researchers using synthetic data must (1) clearly specify which parts of the data are synthetic; (2) explicitly label the synthetic data; (3) describe the methods used to generate the data and (4) explain how and why the synthetic data was utilized.

6. Conclusion

AI presents numerous opportunities, but its use requires a responsible and ethical approach in both pedagogical and research environments. At UM FEB, all users are expected to employ AI in a manner that promotes critical thinking, upholds academic integrity, and respects privacy and intellectual property rules.

References and additional readings:

- AI Guidance, Harvard University, accessed on October 21, 2024, at: https://oue.fas.harvard.edu/ai-guidance
- Dwivedi, Y.K. (2023). Opinion Paper: "So what if ChatGPT wrote it?" Multidisciplinary perspectives on opportunities, challenges and implications of generative conversational AI for research, practice, and policy. *International Journal of Information Management, 71* (2023) 102642. Accessed on October 20, 2024, at: https://www.sciencedirect.com/science/article/pii/S0268401223000233
- Ethical AI for Teaching and Learning, Cornell University, accessed on October 20, 2024, at: https://teaching.cornell.edu/generative-artificial-intelligence/ethical-ai-teaching-and-learning
- Policy Framework Proposal for Responsible and Ethical Use of Generative AI, Maastricht University, accessed on October 20, 2024, at: https://www.maastrichtuniversity.nl/file/policy-framework-proposal-responsible-and-ethical-use-generative-aiucm20230706-lbepdf
- Resnik, D. B., & Hosseini, M. (2024). The ethics of using artificial intelligence in scientific research: new guidance needed for a new tool. *AI and Ethics,* 1-23. Accessed on October 20, 2024, at: https://link.springer.com/article/10.1007/s43681-024-00493-8
- European Commission (2024). Living guidelines on the responsible use of generative AI in research. Brussels: European Commission. Accessed on October 20, 2024, at: https://research-and-innovation.ec.europa.eu/document/2b6cf7e5-36ac-41cb-aab5-0d32050143dc_en

The guidelines were approved by the UM FEB Senate in its session on November 22, 2024.

Maribor, November 22, 2024.